EEE407 Tutorial 3a

1. List down the four strategic thrusts of National Energy Efficiency Action Plan (2016 – 2025).
2. When is a company required to appoint a Registered Electrical Energy Manager (REEM)?
3. What are the functions and duties of a Registered Electrical Energy Manager (REEM)?
4. What electrical appliances are enforced to fulfil standards, criteria, and labelling under Minimum Energy Performance Standard (MEPS)?
5. Evaluate energy management performance for ROSA SDN BHD based on information given in the Appendix A. Use the AEMAS Energy Management Matrix shown in Appendix B for your reference.

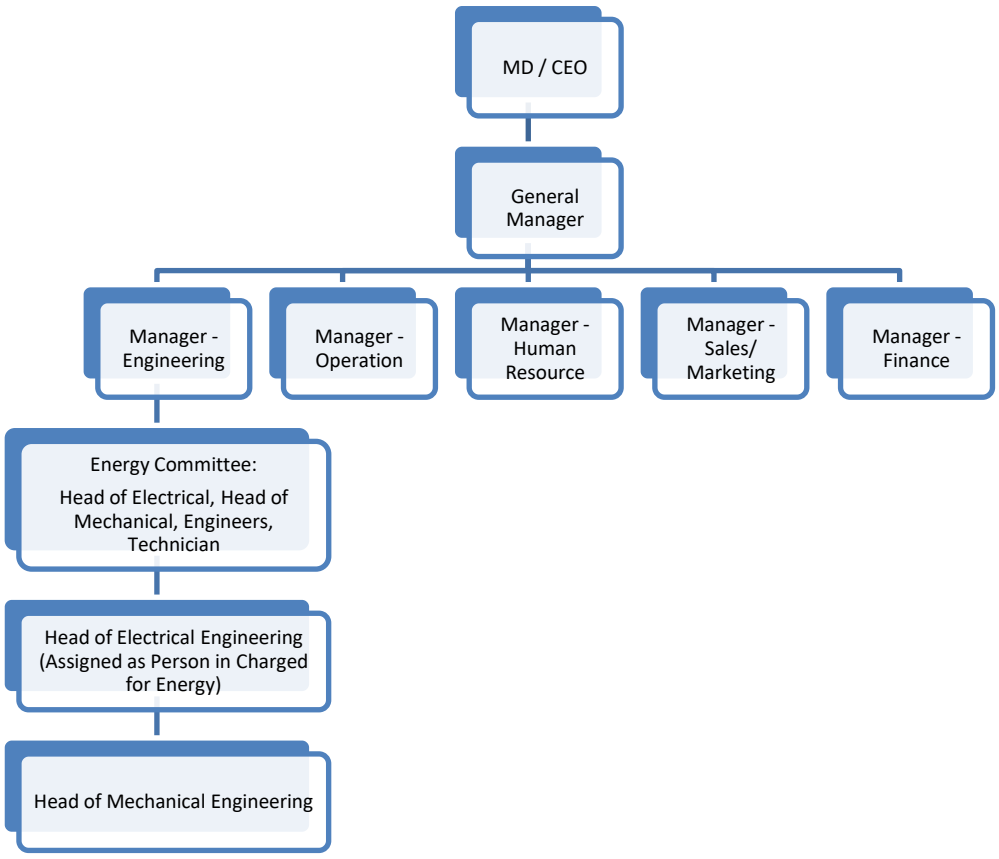
APPENDIX A: ROSA SDN BHD INFORMATION SHEET

1. Energy Policy Statement

ROSA is fully committed to making efficient use of energy at its facilities and for all its activities. We believe in preserving natural resources, reducing atmospheric emissions and helping to mitigate the effects of climate change. The objectives of our Energy Policy are:

a) Identify and evaluate the energy cost of our services and operations and seek either to control or minimize those causing inefficient energy usage, whilst retaining service performance, quality and safety.
b) Set annual targets for continuous improvement, provide adequate resources and implement energy management programs to address significant aspects.
c) Periodically report and review energy performance including targets, improvement activities and compliance programs.
d) Educate all managements and employees via regular energy trainings, so they understand their roles and responsibilities are given in appropriate skills, competencies and involvement, and have good energy awareness.
e) Work with all parties including contractors, suppliers, employees, customers, government agencies, business partners and others to promote energy effectiveness, increase knowledge and disseminate best practice.

2. Company Structure



3. Reporting System

No.	Report receiver	Prepared by	Details	Frequency
1	MD / CEO	-	-	-
2	General Manager	-	-	-
3	Operation Manager	-	-	-
4	Engineering Manager	-	-	-
5	Human Resource Manager	-	-	-
6	Finance Manager	-	-	-
7	Marketing / Sales Manager	-	-	-
8	Head of Electrical Engineering	Technician	Energy consumption data Machine Operation data	Weekly
9	Head of Mechanical Engineering	-	-	-

4. Human Resource Information

Proficiency level of key staffs in the organization

No.	Key Staffs	Authority	Management Skills	Technical Skills	Finance Skills	Communication Skills
1	MD/CEO	H	H	L	M	M
2	General Manager	H	H	L	M	H
3	Engineering Manager	M	H	H	L	M
4	HR Manager	M	M	L	M	H
5	Finance Manager	M	M	L	H	M
6	Marketing / Sales Manager	M	M	L	M	H
7	Engineers	L	M	H	L	M
8	Technicians	L	L	H	L	L

H = High, M = Medium, L = Low

5. Training Activities

No.	Activity	Attendance	Department	Year
1	Quality service training for operation staffs	40 staffs	Operation	2008
2	Safety training	120 staffs	Operation, Engineering	2008
3	Awareness training on ISO 9001:2008	30 staffs	Management team, Engineers, Officers	2009
4	Preventive maintenance training for engineering	50 staffs	Engineering	2009
5	Training on 5S	100 staffs	HR, Finance, Engineering	2010
6	Awareness training on ISO14001	100 staffs	Operation, Engineering	2010
7	Safety training	50 staffs	Operation, Engineering	2011
8	Awareness training on implementation of energy management system	30 staffs	Management team, Engineers, Officers	2011

6. Energy Related Investments

No.	Project Details	Amount (RM)	Payback Period (YR)	Year of Invest
1	Modify operation line - to prepare new service	5,000,000	N/A	2008
2	Improve quality of operation line – to improve service quality	1,200,000	2.1	2009
3	Consult specialist for ISO9001:2008 – to implement of ISO9001:2008	150,000	N/A	2009
4	Replace new lighting system at operation area – to use energy saving lamp	600,000	2.3	2010
5	Consult specialist for ISO14001 – to implement of ISO14001	120,000	N/A	2010
6	Replace new highly efficient motor at operation line – to use energy saving motor	800,000	3.2	2011
7	Install monitoring system – to monitor energy consumption	120,000	3.5	2011

APPENDIX B: AEMAS ENERGY MANAGEMENT MATRIX

Level	Energy policy	Organization	Motivation	Information system	Marketing	Investment
4	Energy policy, action plan and regular review, have commitment of top management as part of an environmental strategy	Energy management has been fully integrated into management structure. Clear delegation of responsibility for energy consumption	Formal and informal channels of communication regularly exploited by energy manager and energy staff at all levels	Comprehensive system sets targets, monitors consumption, identified faults, quantifies savings and provides budget tracking	Marketing the value of energy efficiency and the performance of energy management both within and outside the organization	Positive discrimination in favour of green schemes with detailed investment appraisal of all new build and refurbishment opportunities
3	Formal energy policy, but no active commitment from top management	Energy manager accountable to energy committee representing all users, chaired by a member of the managing board	Energy committee used as main channel together with direct contact with major users	M&T reports for individual premises based on sub-metering but savings not reported effectively to users	Program of staff awareness and regular publicity campaigns	Same payback criteria employed as for all other investment
2	Unadopted energy policy set by energy manager or senior department manager	Energy manager in post, reporting to ad-hoc committee but line management and authority are unclear	Contact with major users through ad-hoc committee chaired by senior department manager	Monitoring and targeting report based on supply meter data. Energy unit has ad-hoc involvement in budget setting	Some ad-hoc staff awareness training	Investment using short term payback criteria only
1	An unwritten set of guidelines	Energy management is the part-time responsibility of someone with only limited authority or influence	Informal contacts between engineer and a few users	Cost reporting based on invoice data. Engineer compiles reports for internal use within technical department	Informal contacts used to promote energy efficiency	Only low measures taken
0	No explicit policy	No energy management or any formal delegation of responsibility for energy consumption	No contact with users	No information system. No accounting for energy consumption	No promotion of energy efficiency	No investment in increasing energy efficiency in premises